ECE 4644 Satellite Communications Satellite Internet

Guest Lecture by Tim Pratt

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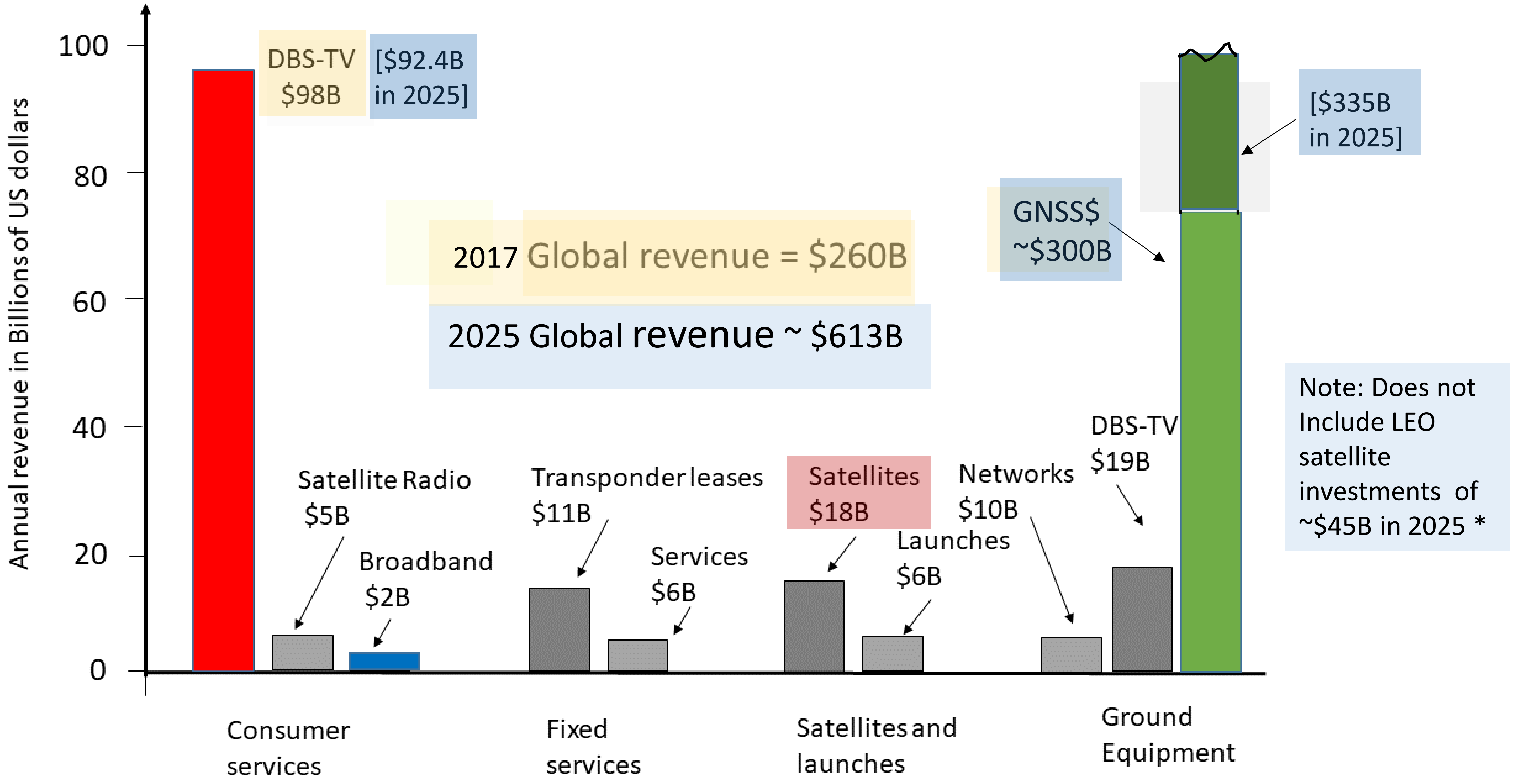
Topics

- Today’s class is based on material in Chapters 10 and 11 of the third edition of *Satellite Communications* by Tim Pratt and Jeremy Allnutt
- Direct Broadcast Satellite Television (DBS-TV)
- Delivered by Geostationary Satellite 1994 through 2025
- Internet Access by satellite 2000 – 2025
- Third generation high capacity GEO satellites with spot beams
- DVB-S and DVB-S2 standards
- Two way links with Adaptive Modulation and Coding
- LEO and VLEO constellations with thousands of satellites

GEO Satellite TV and Internet

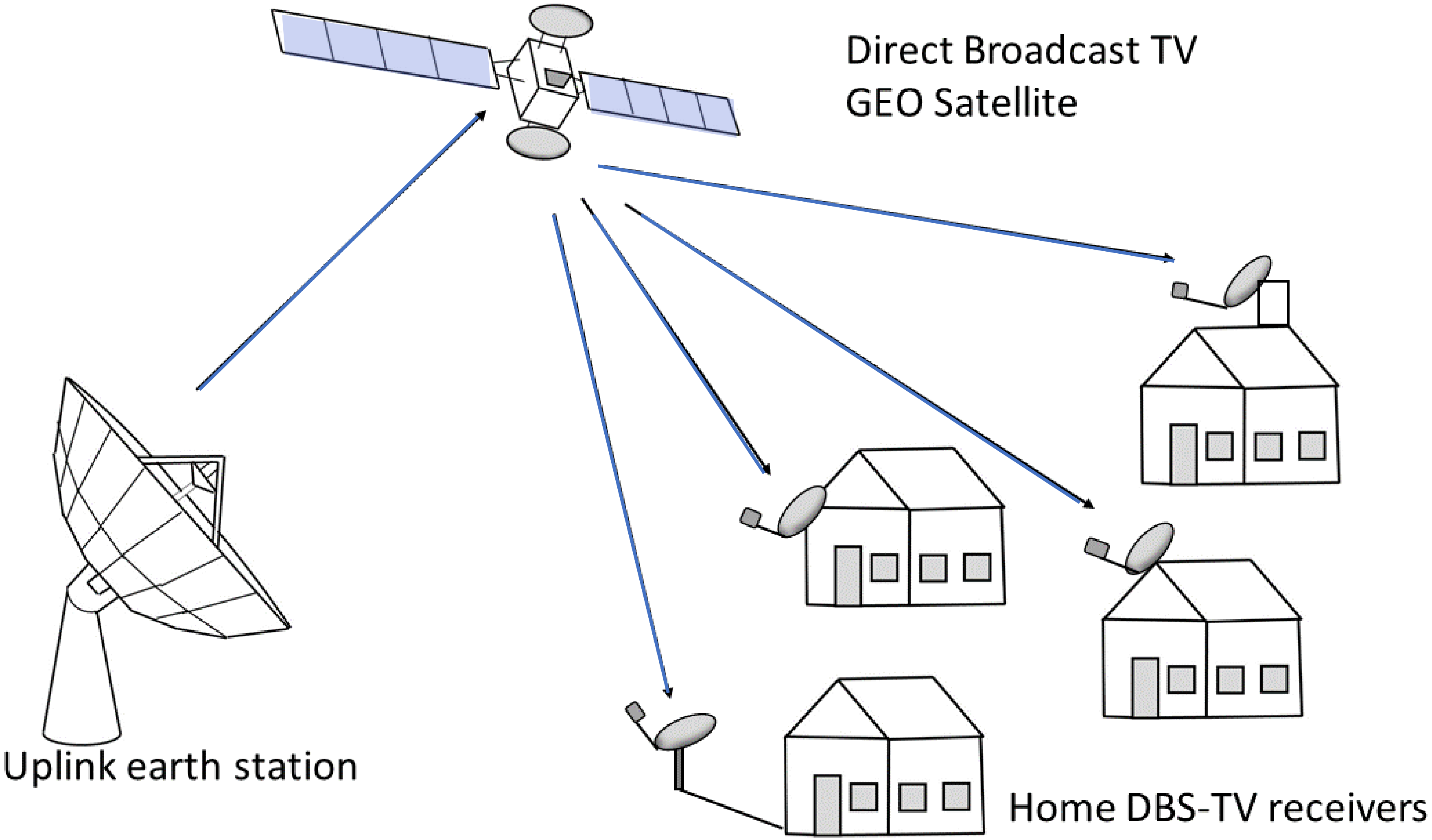
- Distribution of television signals by geostationary satellites produced nearly half of the global revenue from all satellite services in 2017
- Directv and Dish Network are the major providers in the US
- Global revenue from TV distribution was $98 B in 2017
- Internet access revenue was $2 B
- DBS-TV has been losing customers to other sources since 2014



Global revenue from all satellite services in 2017 [updated to 2025]

DBS TV and Internet Access

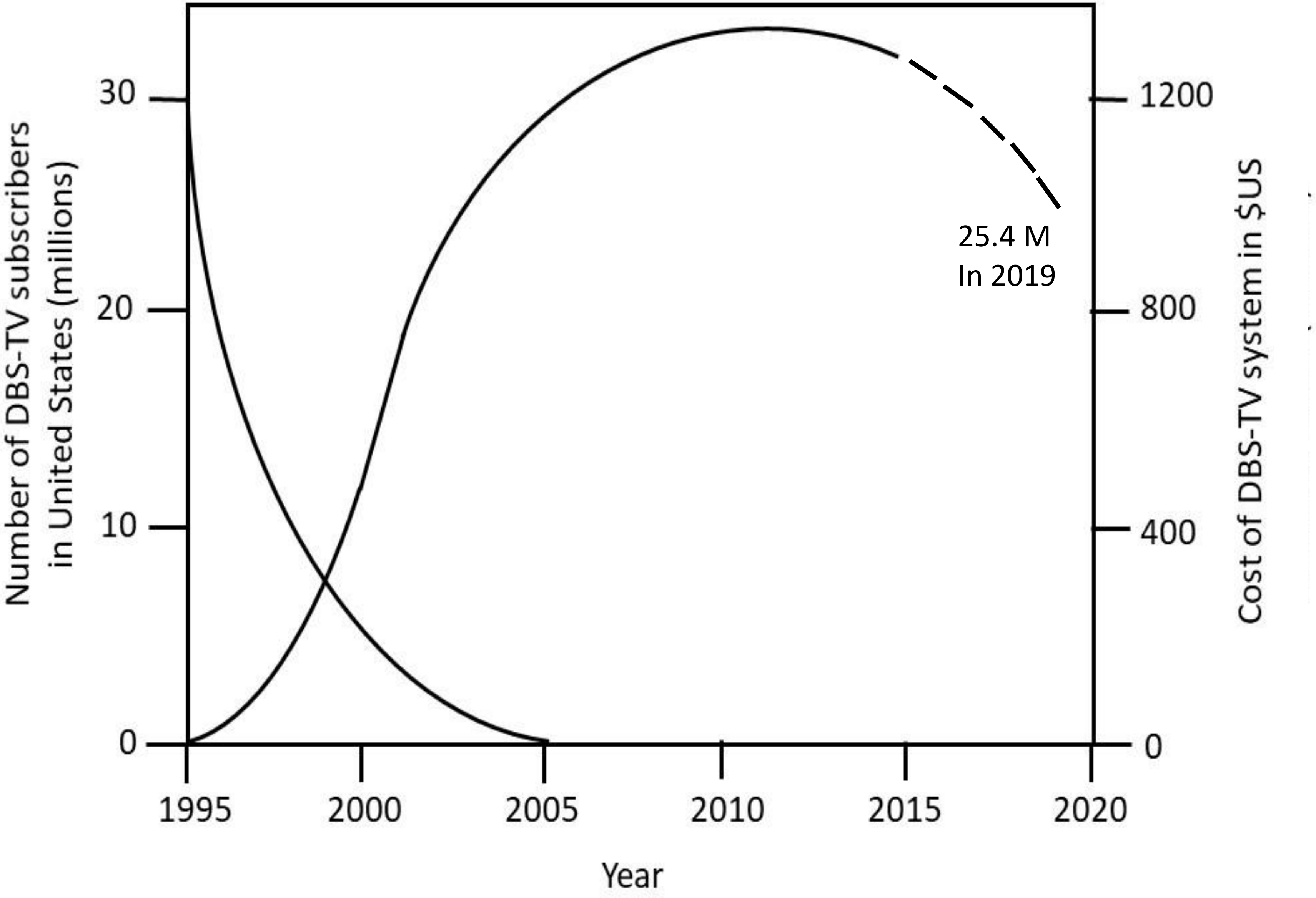
- Directv began transmissions in 1994, Dish in 1996, each with a single satellite
- Standard receiving antenna was 0.5 m (20 inch) circular parabolic dish
- Flatter dish has elliptical shape and can receive from several satellites
- DBS-TV was allocated Ku-band frequencies 12.2 to 12.7 GHz
- All transmissions were digital using DVB-S standard with QPSK modulation, half rate FEC, and MPEG 2 video compression
- Spot beams on later satellites can use 8-PSK with ¾ rate FEC to increase bit rate
- Used with MPEG-4 compression for HD TV transmissions



Direct Broadcast satellite TV system at Ku-band

1996 Directv receiving antenna with single feed

2012 Dish Network receiving antenna with three feeds



DBS TV customer base in USA   End 2022 subscriber number was 20.5M

Early Satellite Internet

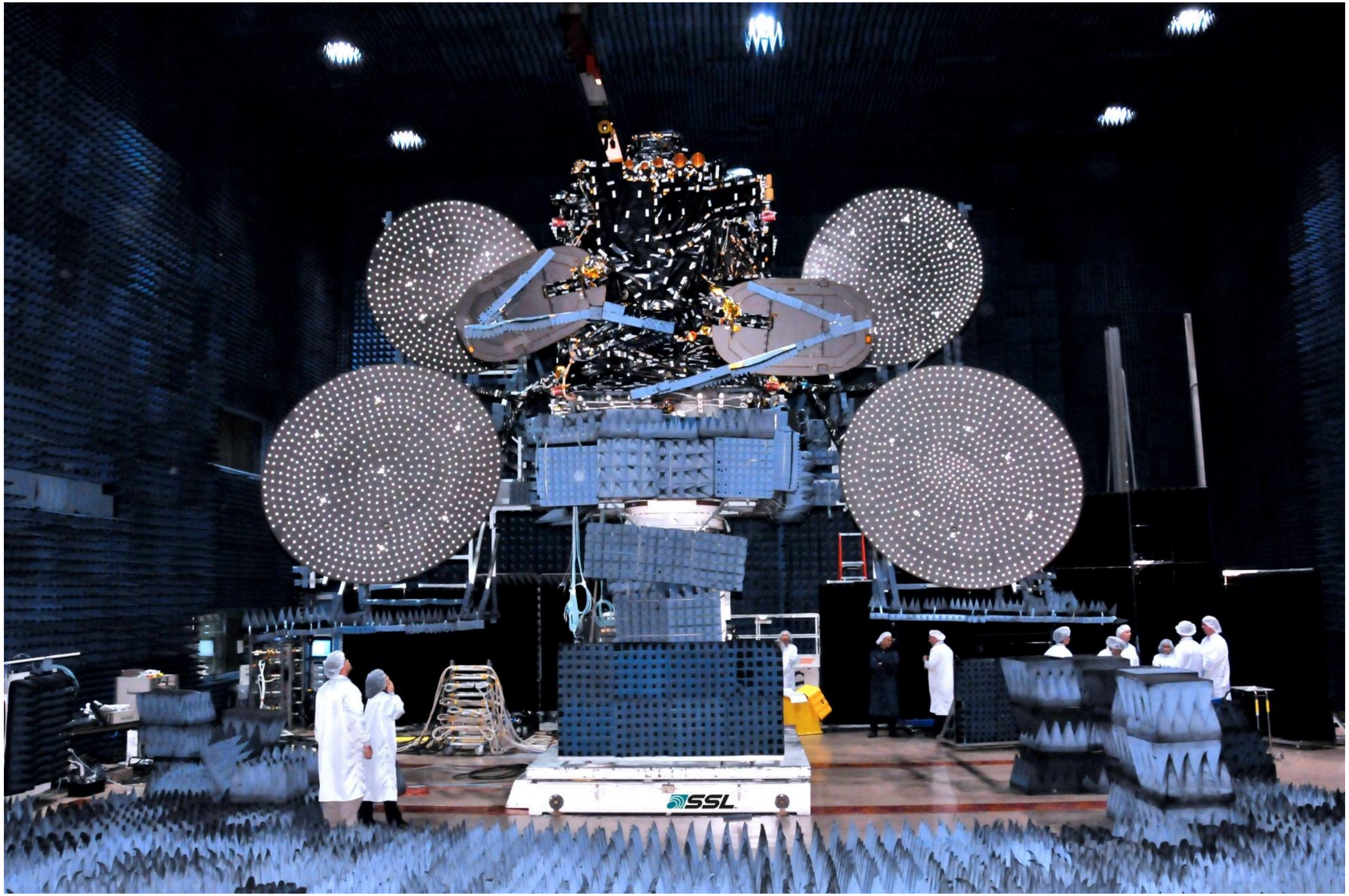
- Directv and Dish offered Internet access using transponders on their DBS-TV satellites.   Total capacity of one GEO satellite was less than 5 Gbps.
- Companies oversold the service - contention ratios of 5000 to 1 were typical
- Contention ratio is number of users sharing a single channel
- Advertised data rate was typically *up to* 1 Mbps down, 128 kbps up
- Throughput in evening fell to less than 10 kbps

Third generation GEO DBS-TV satellites

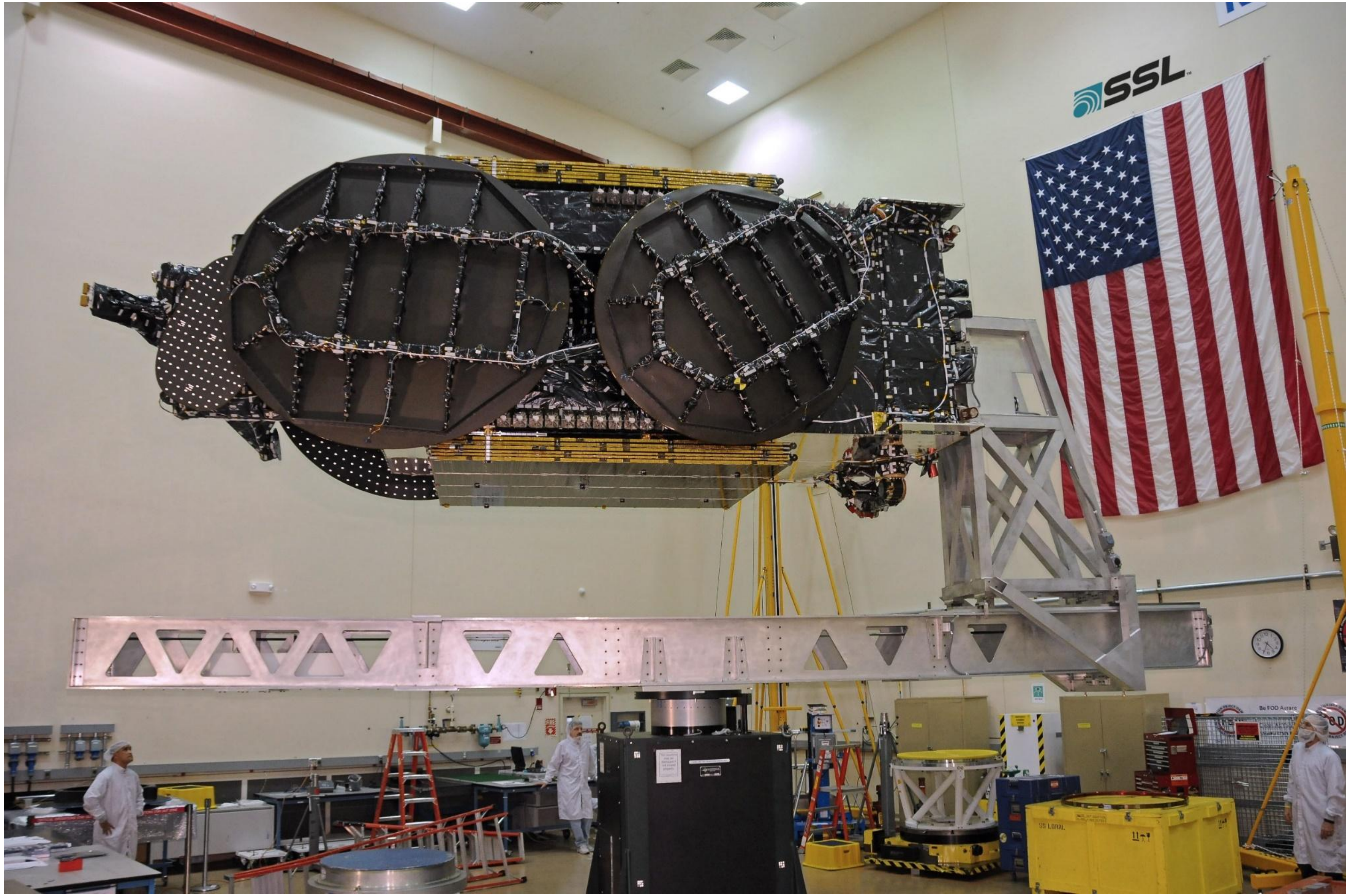
- Third generation DBS-TV satellites use Ku- and Ka- bands with many spot beams to increase capacity to 100 Gbps and above
- In 2011 ViaSat launched ViaSat 1 with 140 Gbps capacity using Ka-band
- 1.5 GHz bandwidth with 72 spot beams and multiple frequency reuse
- Intended for DBS-TV and Internet access
- Hughes Networks sells Internet access on Jupiter satellite with speeds *up to* 25 Mbps on downlink and 5 Mbps uplink
- Typical cost for a large GEO satellite is $100 M+ with additional $50 M for launch

Artist’s rendition of Viasat 1 in orbit © Viasat 2018

Artist’s rendition of Intelsat 35e © Intelsat 2018



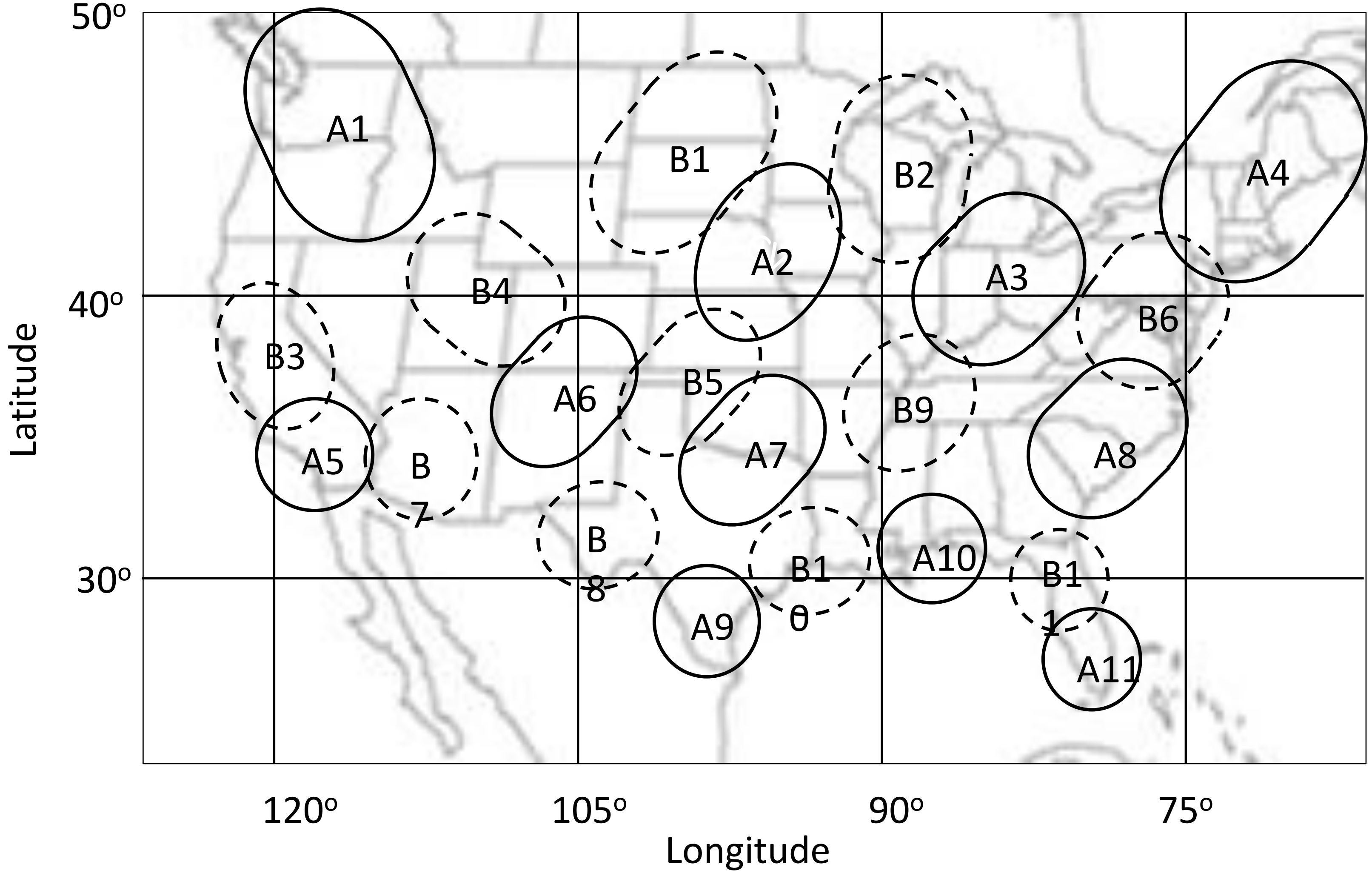
Large GEO DBS-TV satellite built by SSL ©SSL 2018



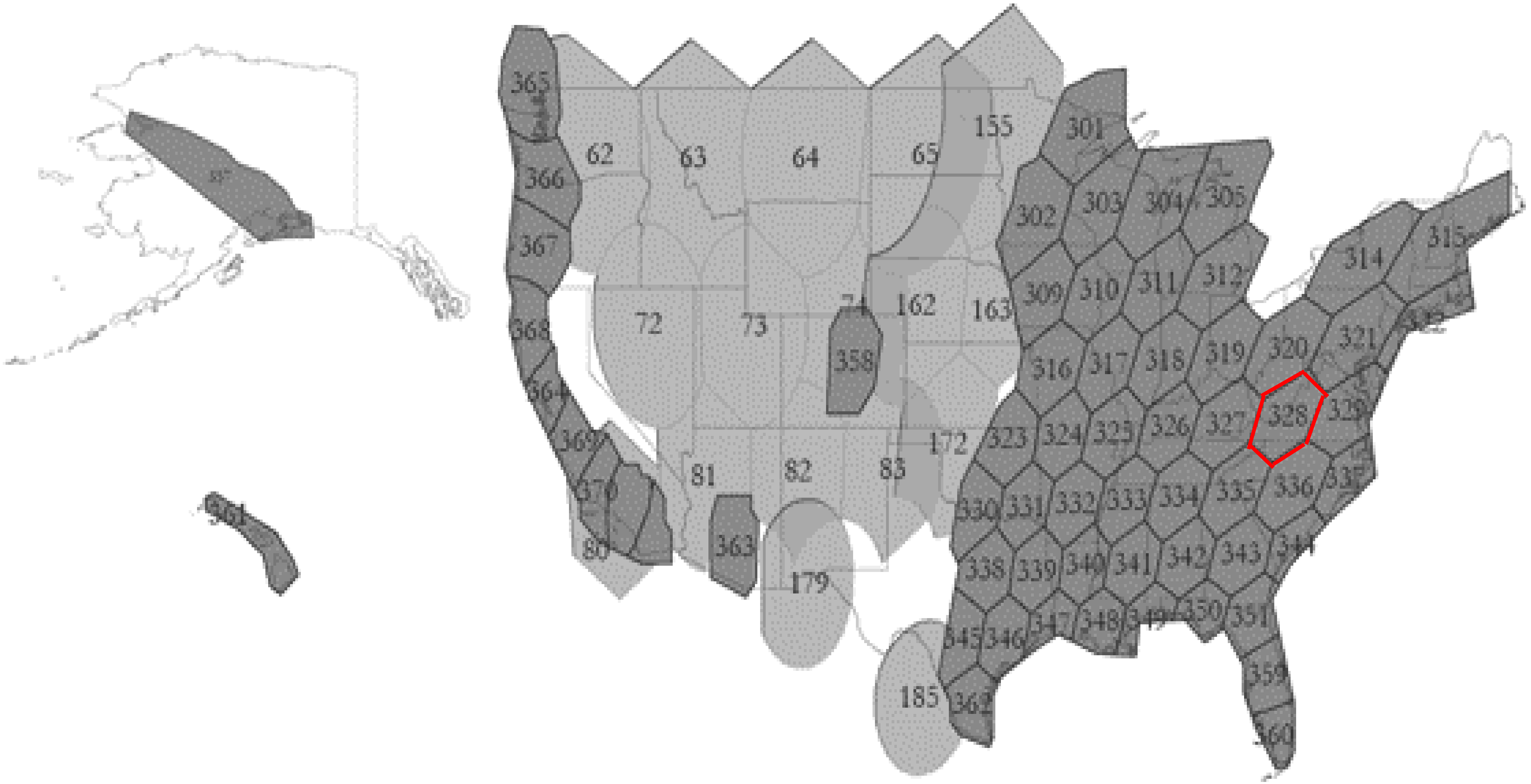
SSL DBS-TV satellite folded for launch ©SSL 2018

2005 era Ku band Internet access customer terminal

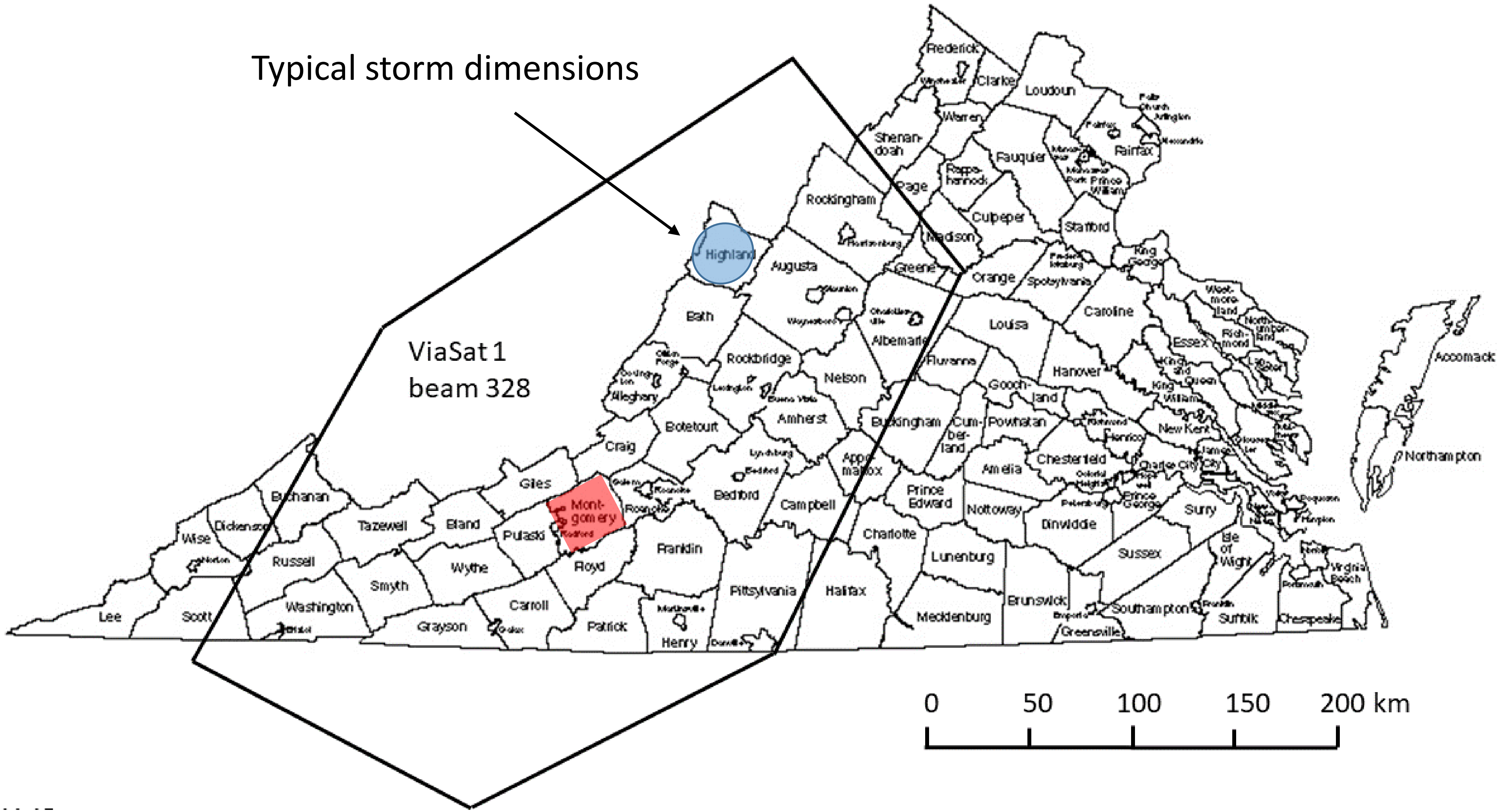
Elliptical feed of Ku band Internet access terminal



Typical Ku-band spot beams from a GEO satellite
A and B series beams are in different frequency bands

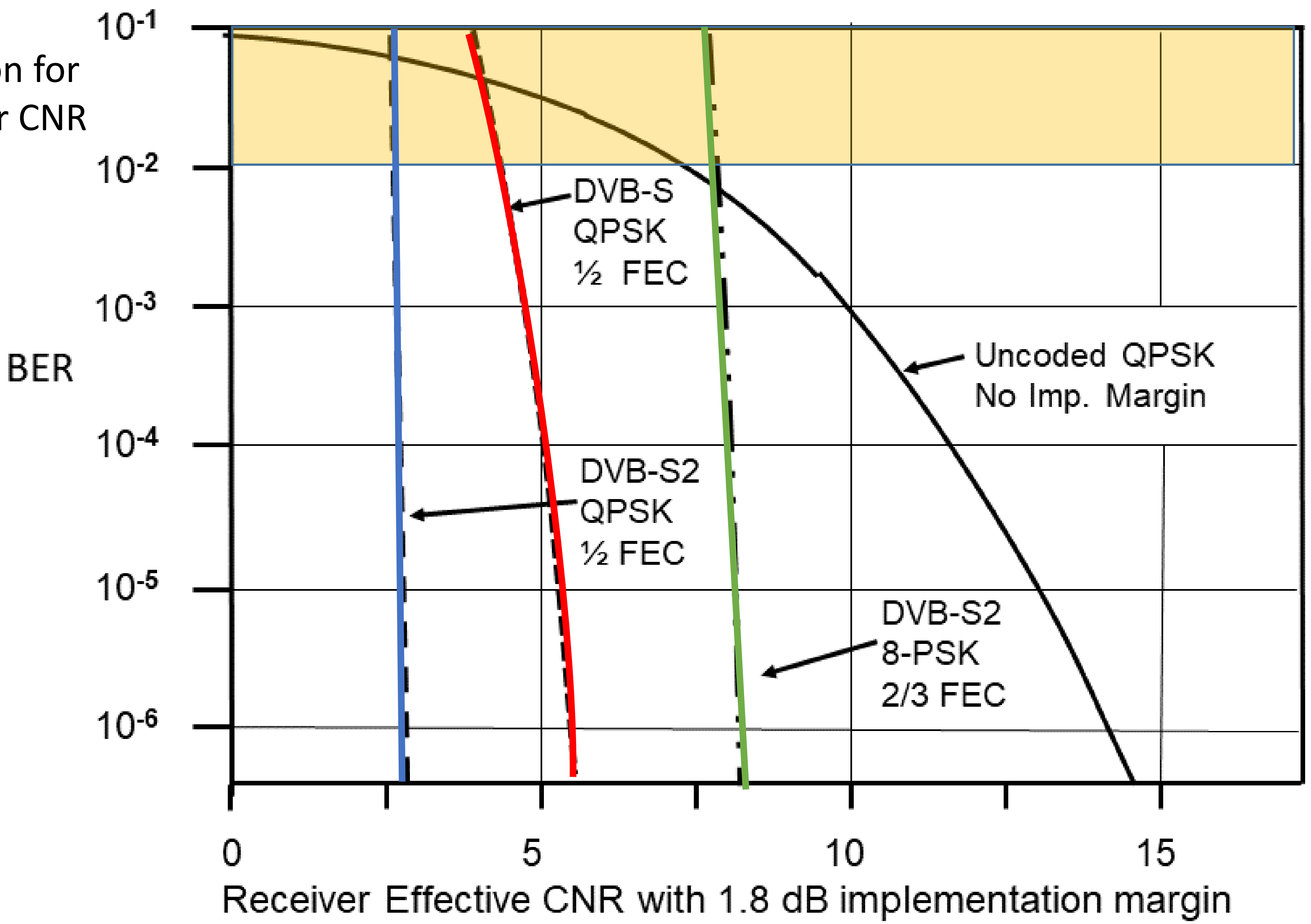


ViaSat 1 spot beam coverage of USA. Beams numbered 3xx are ViaSat 1. Lower numbers are beams from the Wild Blue satellite.



ViaSat 1 beam # 328 centered on Blacksburg

Operating region for DVB-S2 receiver CNR in rain



DVB-S and DVB-S2 BER performance - QEF objective is one bit error per hour

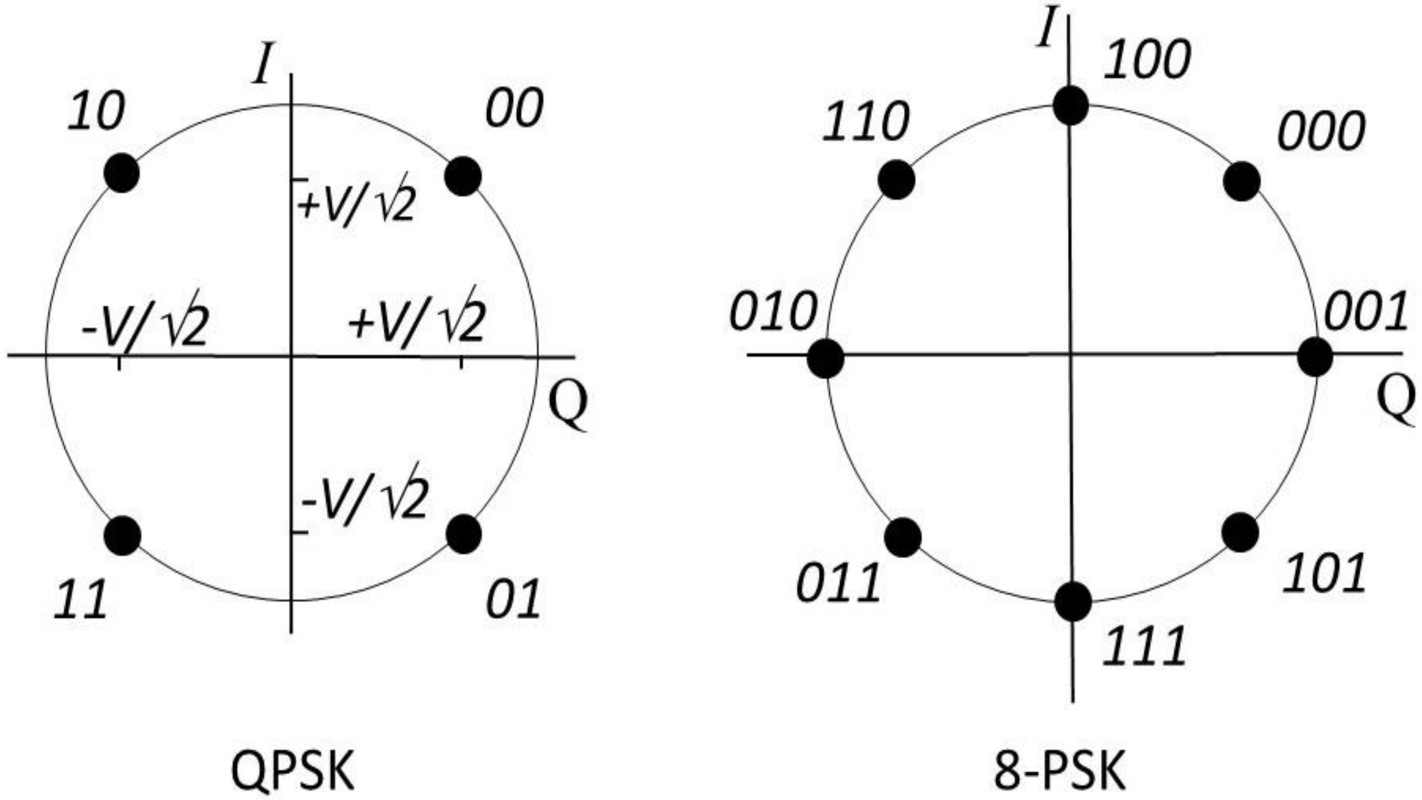
QEF = Quasi Error Free

Internet Access via GEO satellite

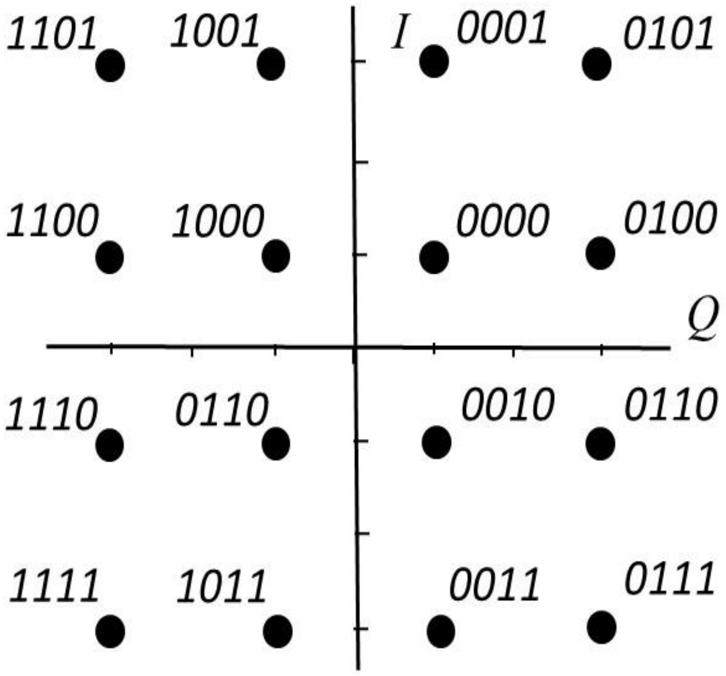
- Two way communication is required for Internet access
- Almost all systems use the DVB-S2 standard
- DVB-S was developed by the European Technical Standards Institute (ETSI) in the 1997 – 2003 time period for DBS-TV transmissions
- Began with QPSK and half rate FEC using two layers of encoding – inner layer of convolutional coding, outer layer of Reed Solomon coding
- Later versions could employ 8-PSK and ¾ rate FEC for higher bit rates
- Typical two way link delay is 700 ms

Internet Access via GEO satellite

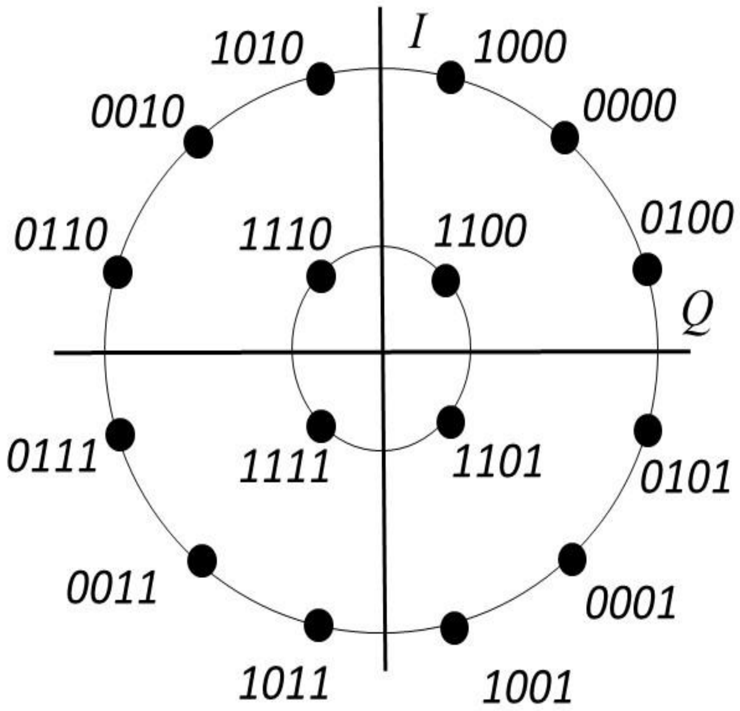
- DVB-S2 was developed by ETSI in 2003 - 2009 time period
- Modulations include 16-APSK, 8-PSK, QPSK, with FEC rates from 9/10 to 1/4
- Transmissions are 188 byte packets of MPEG-4 coded TV, or data
- Packet has Reed-Solomon coding
- 8 or 32 packets are assembled into a frame with BCH and LDPC coding
- Raw error rate on link can be 10^{-1} to 10^{-2}
- One in 10 to one in 100 bytes have errors; corrected by three layer FEC
- After error correction, *Quasi error free* (QEF) objective is BER of 10^{-11}
- Adaptive Modulation and Coding (ModCon) is used on two way links
- Large ICs are available that implement all the features of two way DVB-S2



QPSK and 8-PSK constellations - Constant amplitude



16 QAM



16-APSK

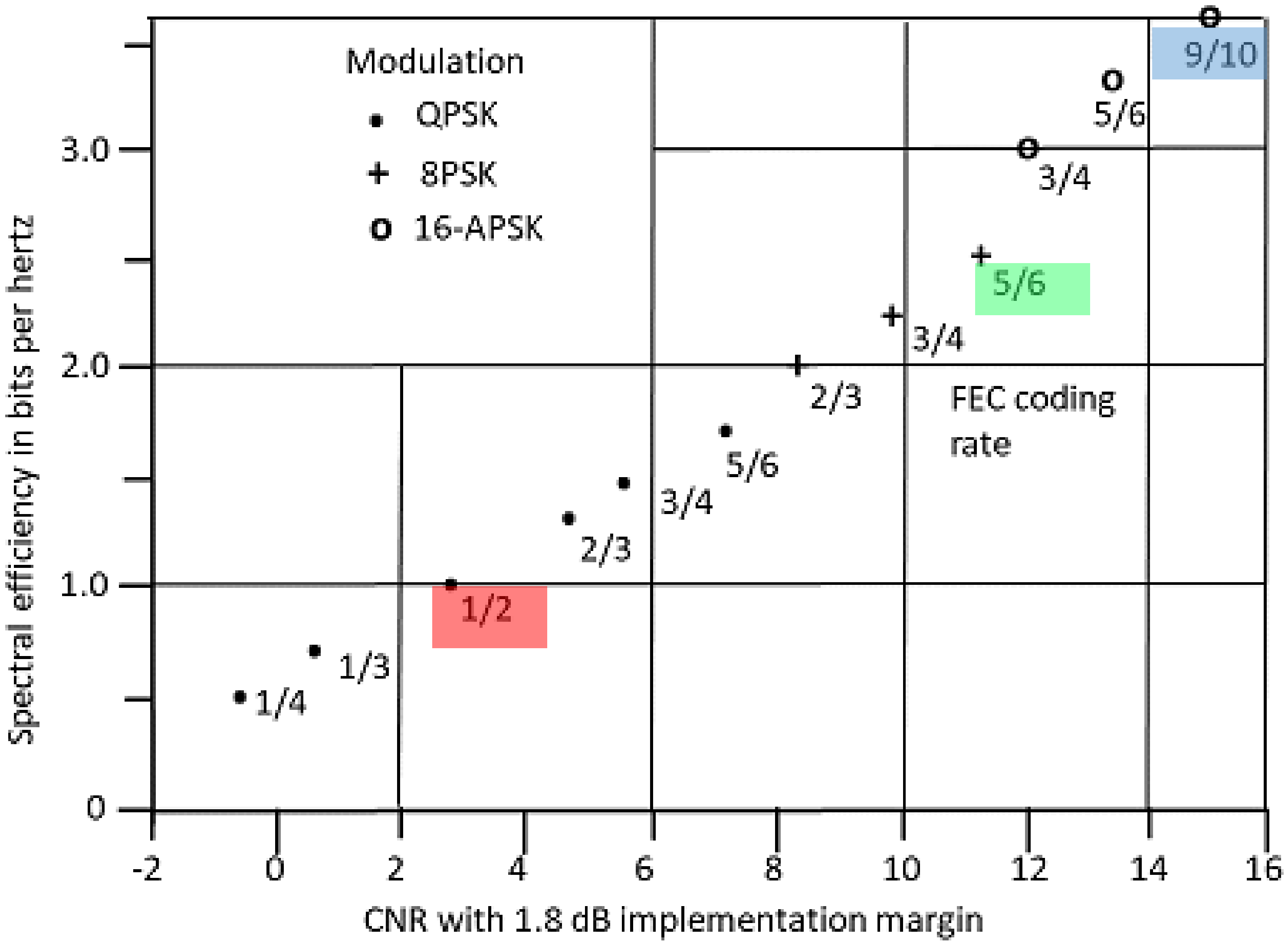
16 QAM and 16-APSK constellations – 16-APSK is used in DVB-S2 standard
Different amplitudes require a linearized transponder

Adaptive Coding and Modulation with DVB-S2

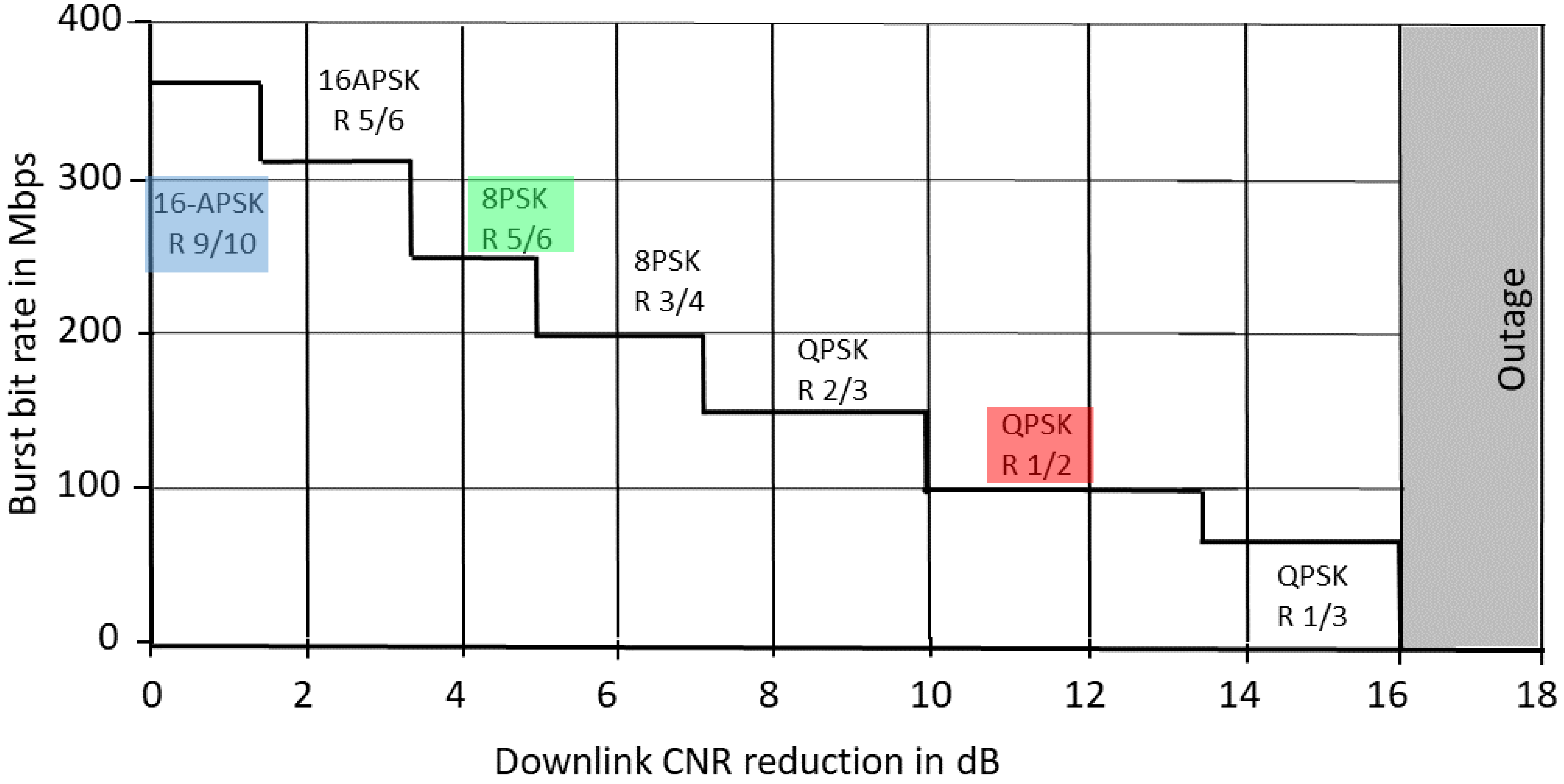
- Ka band links suffer from more rain attenuation than Ku-band links
- Direct broadcast satellite TV using DVB-S has fixed modulation and coding
- For 95% of time, atmospheric attenuation in Ka-band is less than 1 dB
- Doesn’t need half rate FEC for 95% of time
- Two way Internet Access links establish a virtual circuit between the Gateway station and each user
- Adaptive coding and modulation optimizes the bit rate for each user

Adaptive Coding and Modulation with DVB-S2

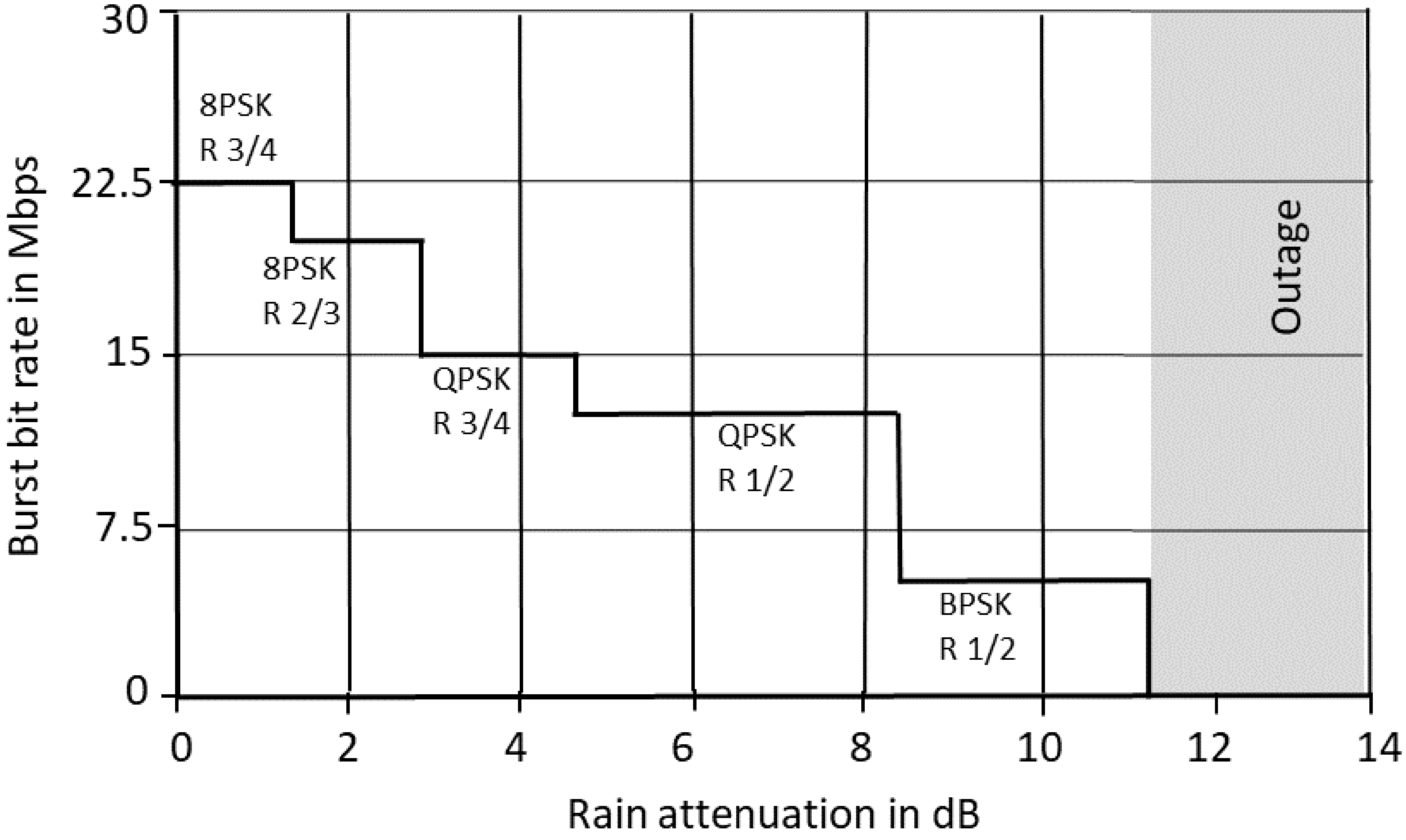
- At the center of a satellite spot beam the CNR in the receiver is maximum
- Adaptive ModCon allows highest order modulation and smallest FEC rate
- Example: Receiver at center of downlink beam has CNR 16 dB
- ModCon is 16 APSK with 9/10 rate FEC, bit rate is 360 Mbps in clear air
- On -3 dB contour of beam CNR is 13 dB, ModCon is 8-PSK, 5/6 rate FEC, 250 Mbps
- With rain attenuation, higher FEC rates are needed and step down to QPSK
- 12 dB reduction in CNR requires QPSK with half rate FEC, 100 Mbps



Bits per hertz for some DVB-S2 ModCons



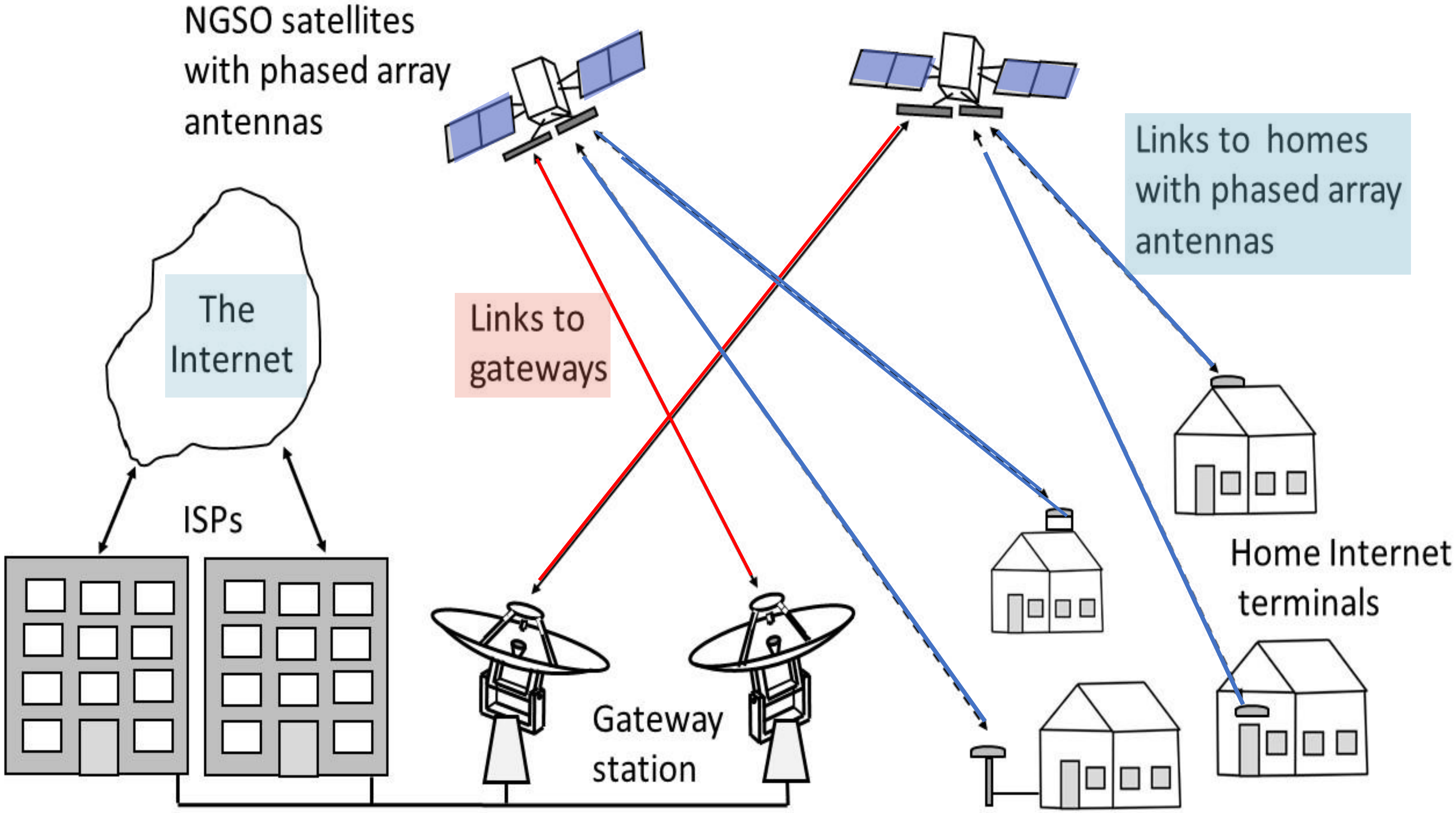
Example of Adaptive Modulation and Coding to combat rain attenuation on downlink



Example of Adaptive Modulation and Coding to combat rain attenuation on uplink

Low Earth Orbit Constellations for Internet Access

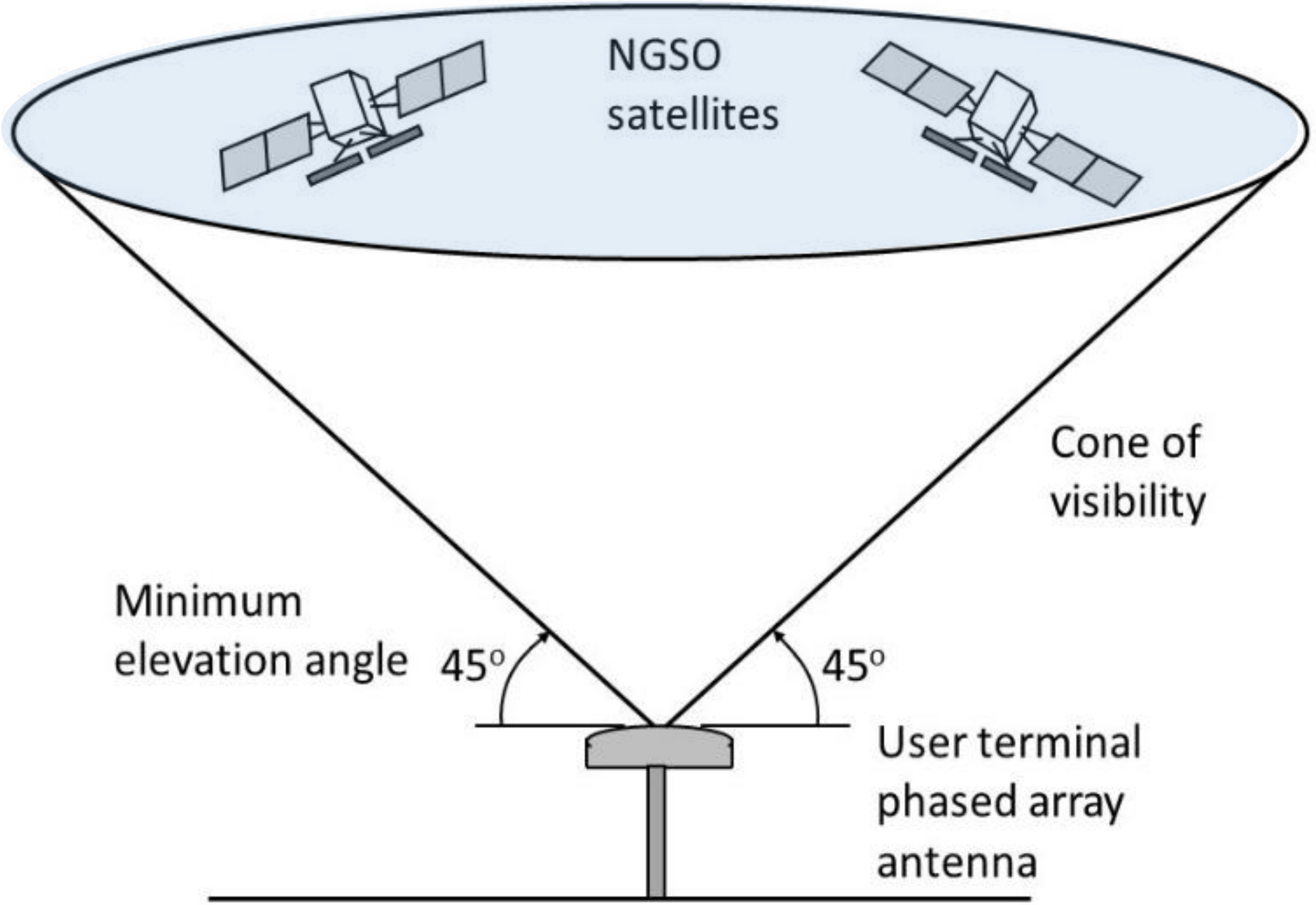
- The dominance of GEO satellites may be coming to an end
- By 2027 there may be more than 40,000 LEO satellites in orbit providing world wide Internet access
- LEO satellite orbit is also known as Non-Geostationary Satellite Orbit - NGSO
- Example: SpaceX V-band constellation of 7500 very low earth orbit (VLEO) satellites at 350 km altitude [SpaceX now proposes 40,000 satellites by 2025]
- 4 GHz bandwidth in 38 – 50 GHz band, 360 Mbps downlink rate, 22 Mbps uplink
- Target: $0.5 M per satellite, $0.5 M per launch … cost for 7500 sat constellation: $7.5 B



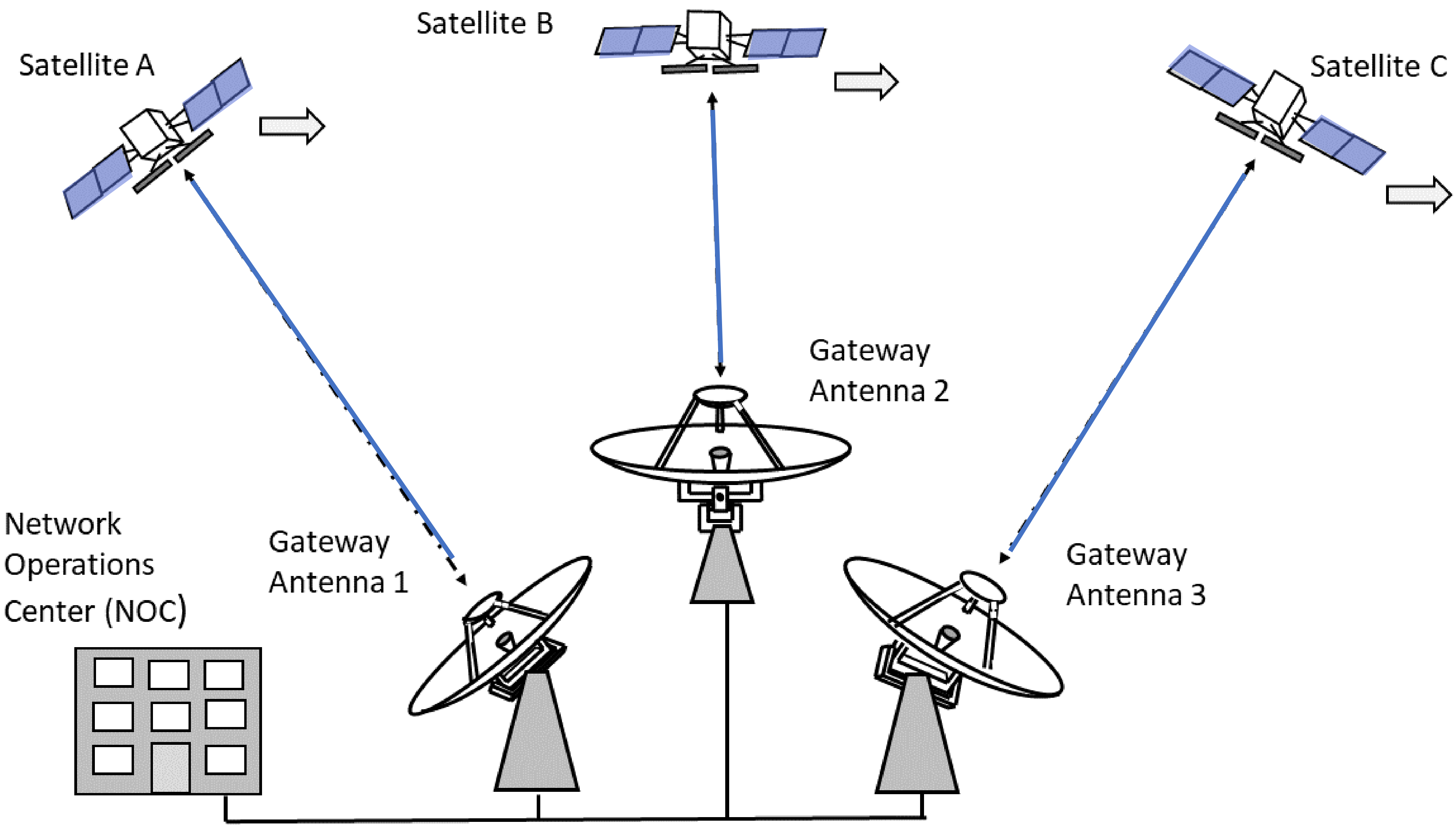
Internet access with Non-Geostationary Orbit (LEO) satellites
Satellites have multiple electronically steered beams

Low Earth Orbit Constellations for Internet Access

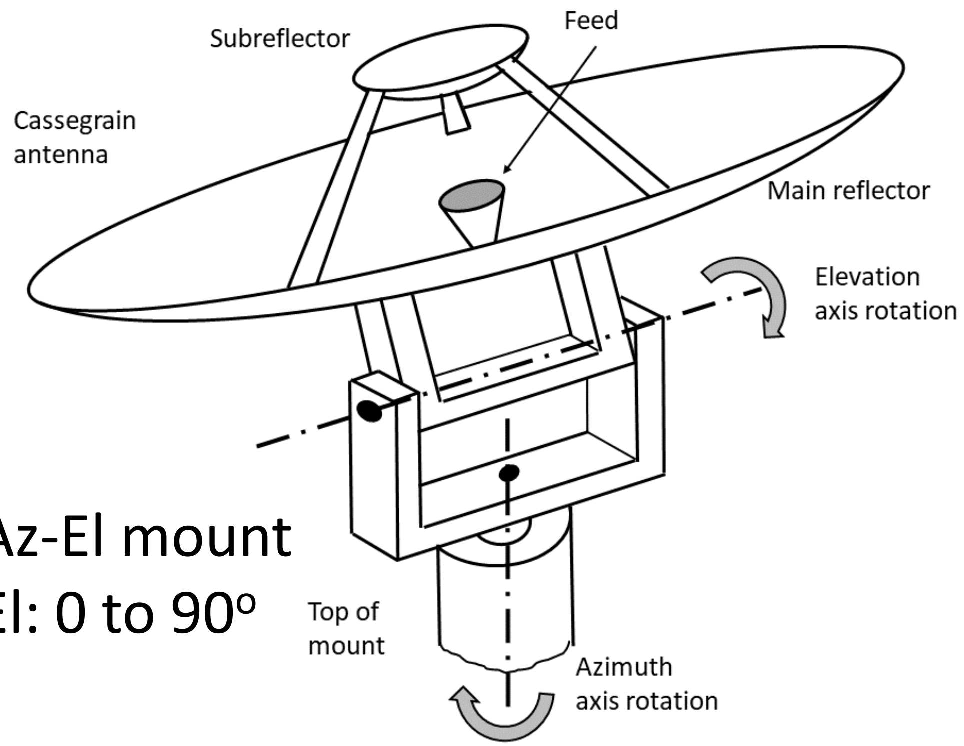
- LEO satellites move across the sky … must be tracked by user
- Phased array antenna is needed at user terminal - at acceptable cost
- Phased array antenna can track $\pm$45 degrees from zenith
- Communication is possible within a *cone of visibility* for maximum of 134 seconds with satellite at 350 km altitude (a VLEO altitude)
- Gateway stations need many tracking antennas on x-y mounts
- Satellite needs large phased array that can switch between users in microseconds



Cone of visibility for phased array user terminal

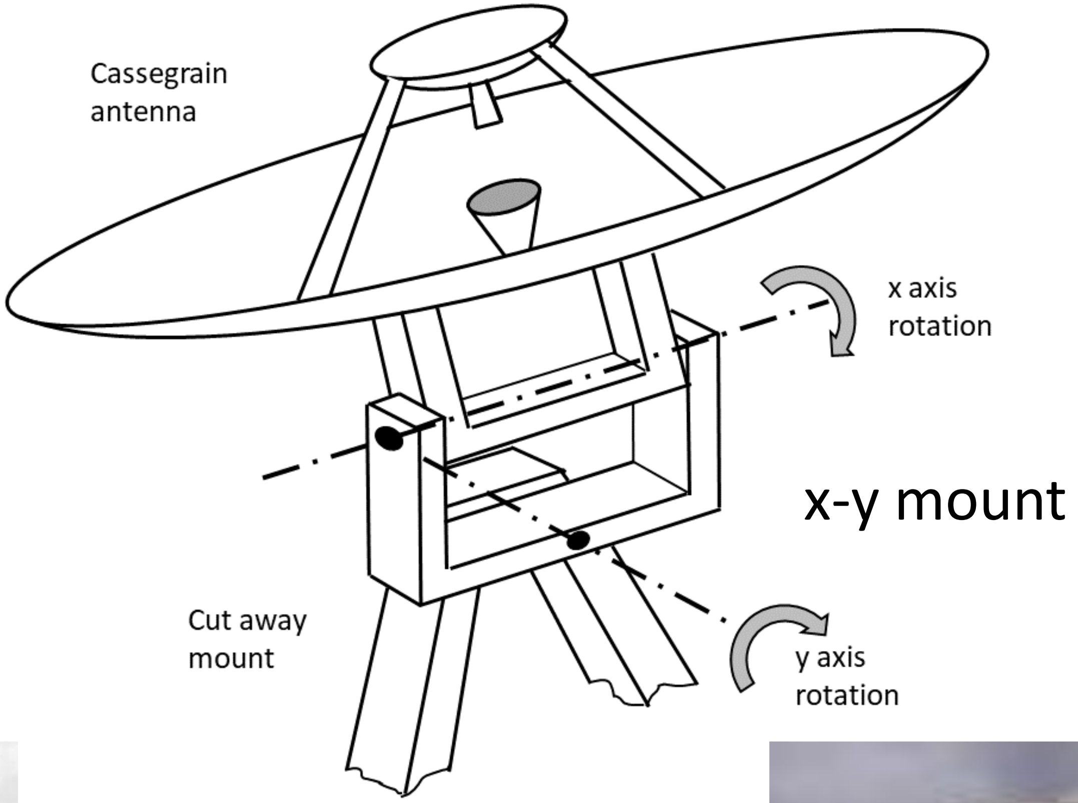


VLEO gateway station. Antennas have x-y mounts to track satellites through zenith



Az-El mount

El: 0 to 90^o

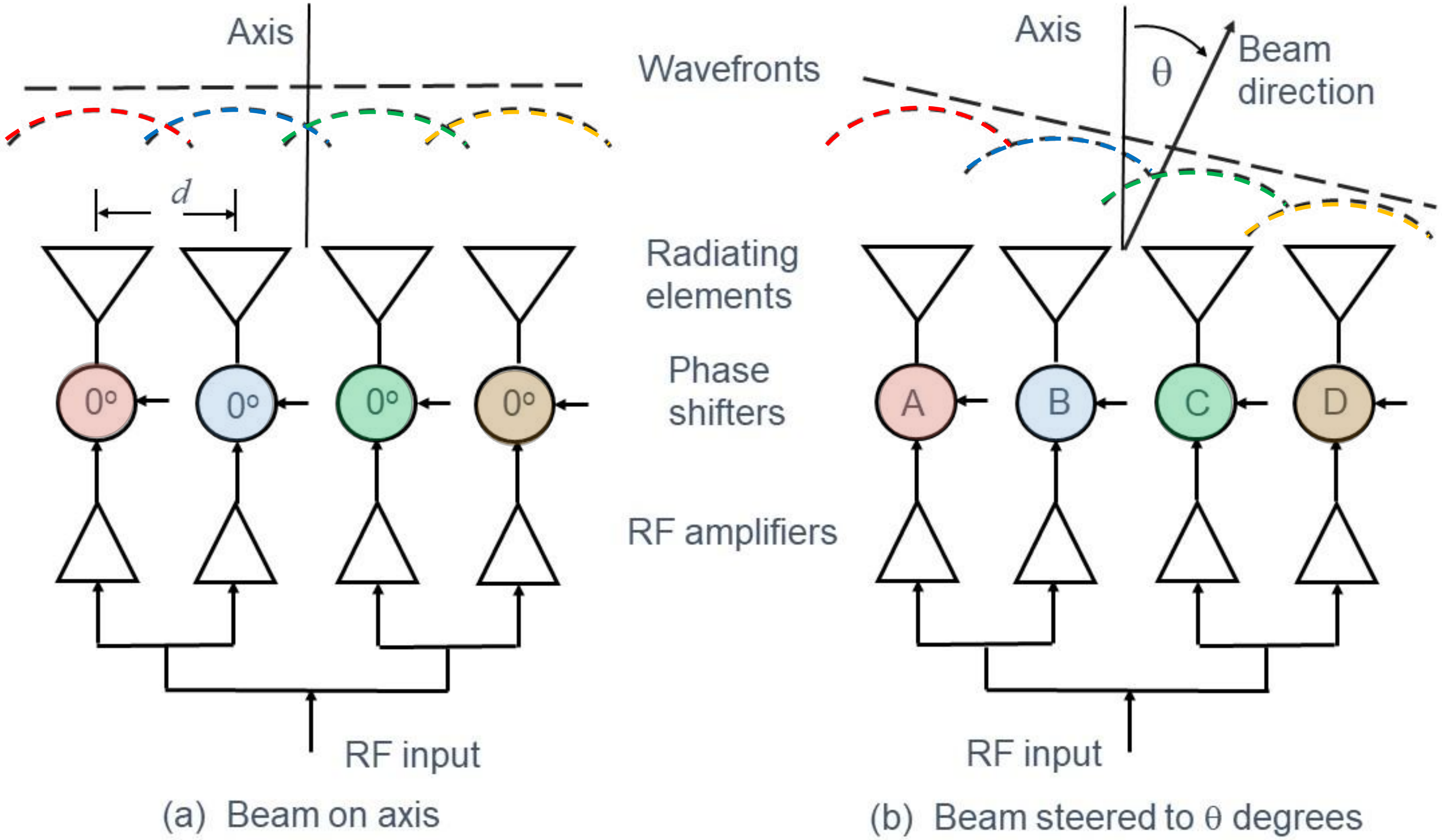


x-y mount

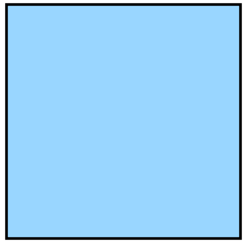
Antenna with x-y mount

Phased Array Antennas

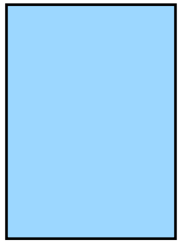
- For more information on antennas, read Appendix B of the class text
- Phased array antennas are made up of many active elements
- Typically, separate transmitting and receiving antennas are used for LEO satellite systems
- Beam pointing is by computer controlled command with repointing times in microseconds
- LEO satellite phased array antenna points at an individual earth station terminal for 1 millisecond
- At 360 Mbps delivers 360 kb
- Beam must return 14 times each second to make downlink bit rate 5 Mbps
- One beam can serve 72 user terminals within cone of visibility – footprint is 700 km wide
- Satellite can have 20 downlink beams at different frequencies across 2.5 GHz bandwidth
- Five visible satellites x 20 beams x 72 users = 7200 users
- With 50:1 contention ratio, system serves 36,000 users in one footprint
- With 40,000 satellites system can serve 2.8 billion users. Hence ”the other 3 billion – O3B”



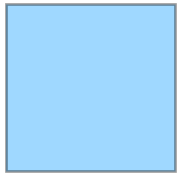
Principle of a phased array antenna.
Beam is steered by successively increasing phase shift
of phase shifters D through A in (b).
Elements are about one half wavelength apart and omni-directional.
Antenna gain is reduced as beam is scanned away from axis.



Array area at normal incidence

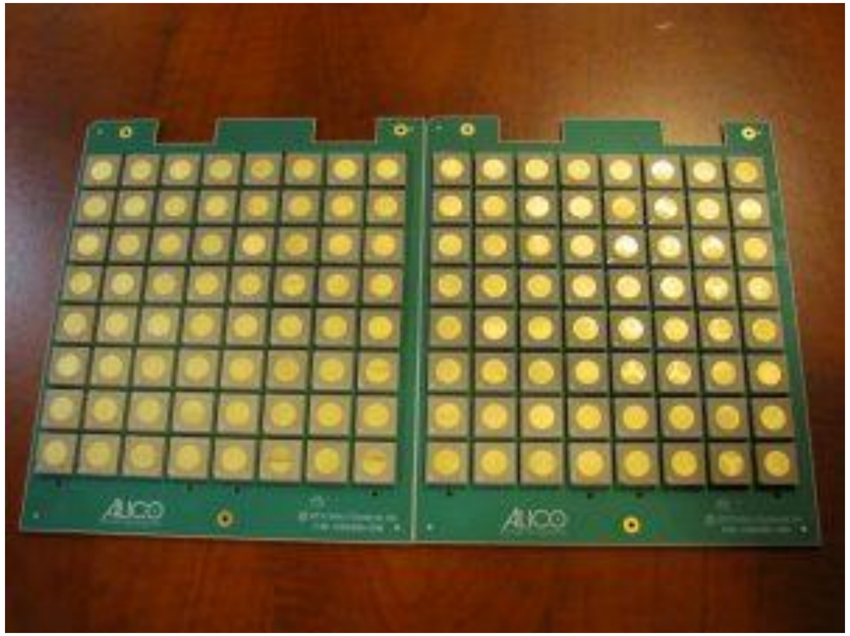


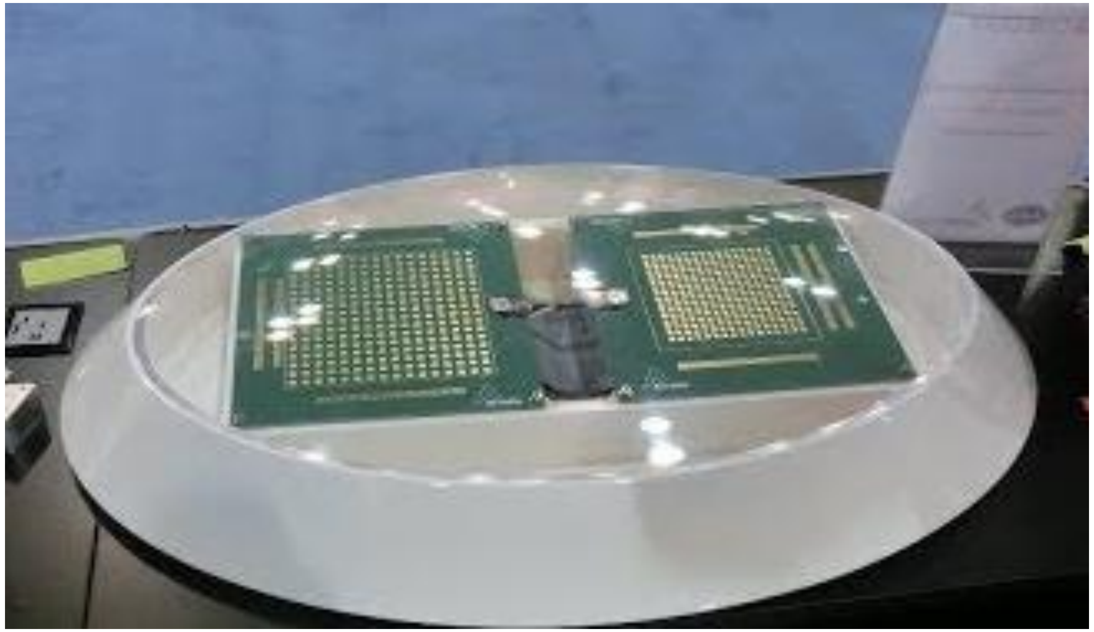
Array area at 45° incidence in one plane

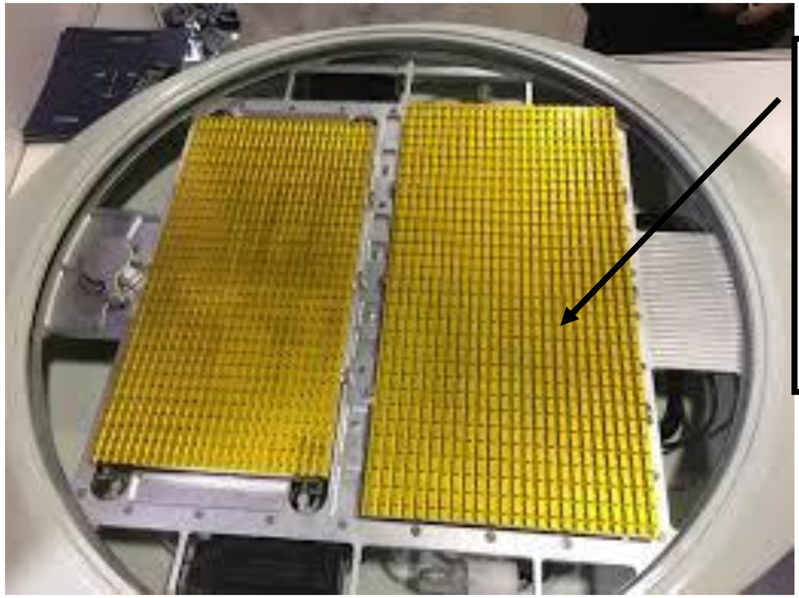


Array area at 45° incidence in both planes.

Antenna gain is reduced by 3 dB

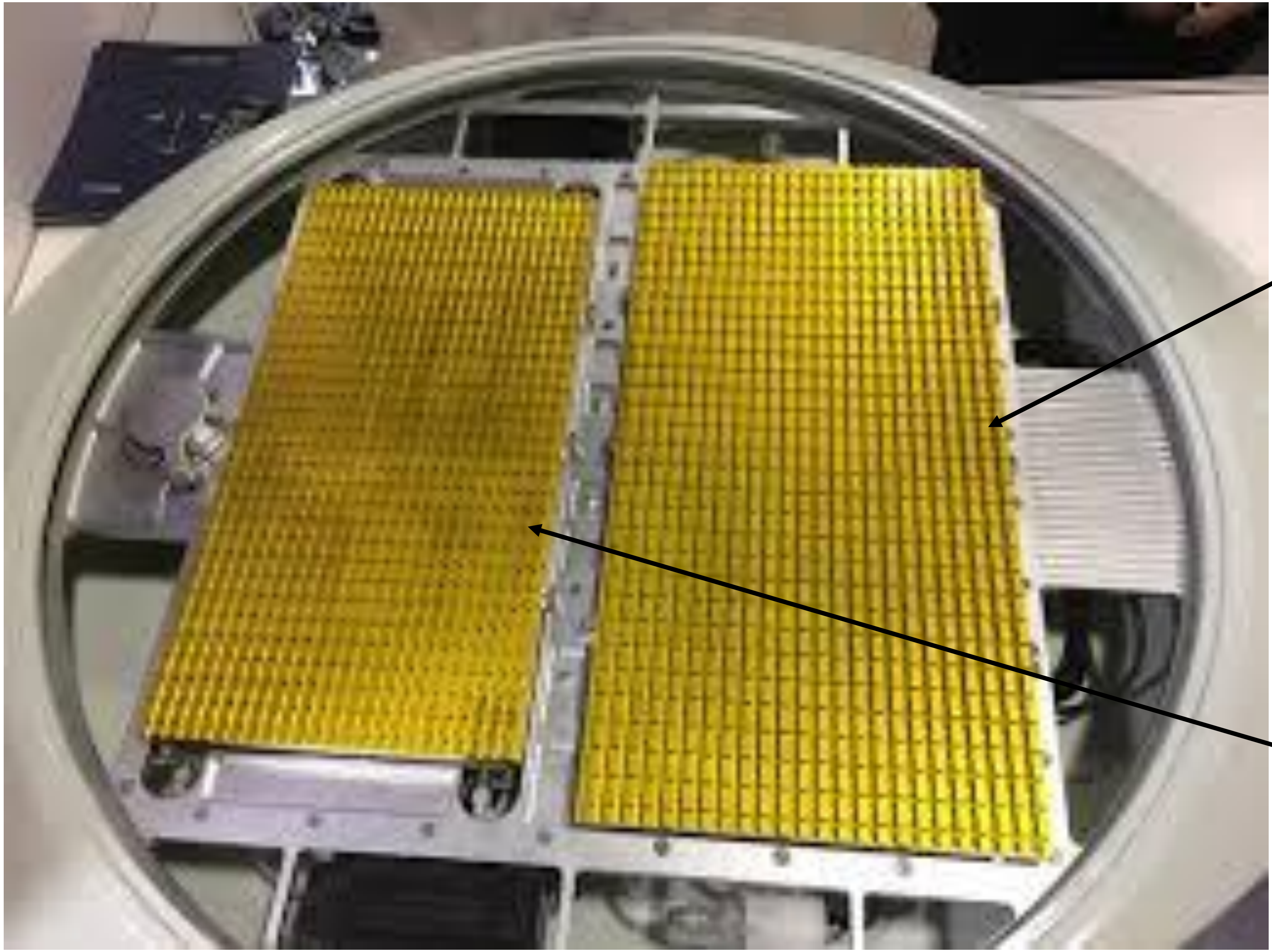






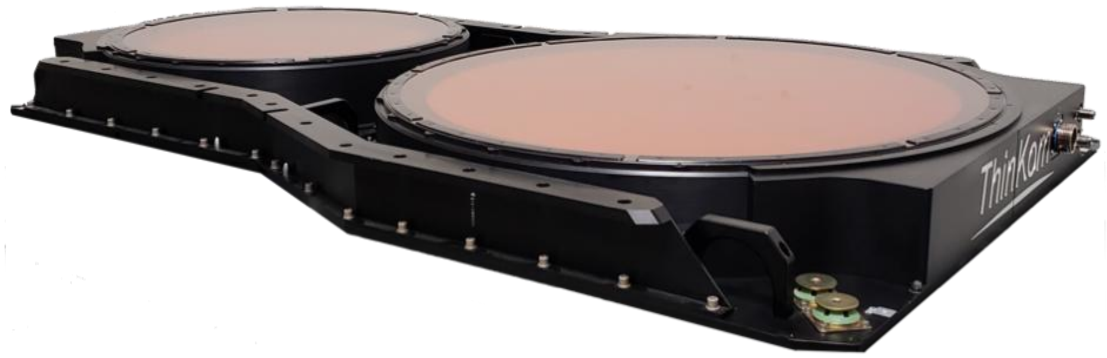
This array appears to have about 23 x 40 = 920 elements. Gain ~ 33 dB

Examples of Ku-band phased array antennas.
Element spacing must be close to one half wavelength to achieve 45 degree scan angle.
For 30 dB gain, with 80% aperture efficiency, area required is 100 λ^2 .
Square array is 10λ on a side, requiring ~20 elements/side.
Hence, the array requires 20 x 20 = 400 elements.



12 GHz Receive array

18 GHz Transmit array



Thinkcom Ka band phased array antenna

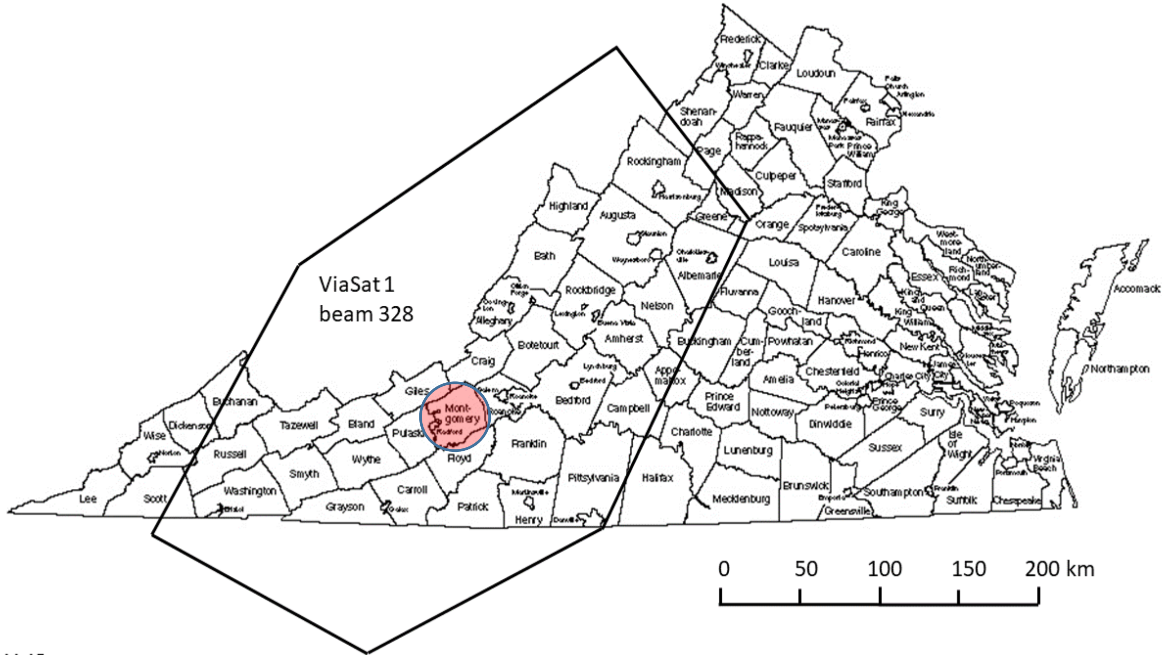
OneWeb Ka band
phased array antenna

Starlink Ku/K band
phased array antenna
Uses mechanical
repositioning

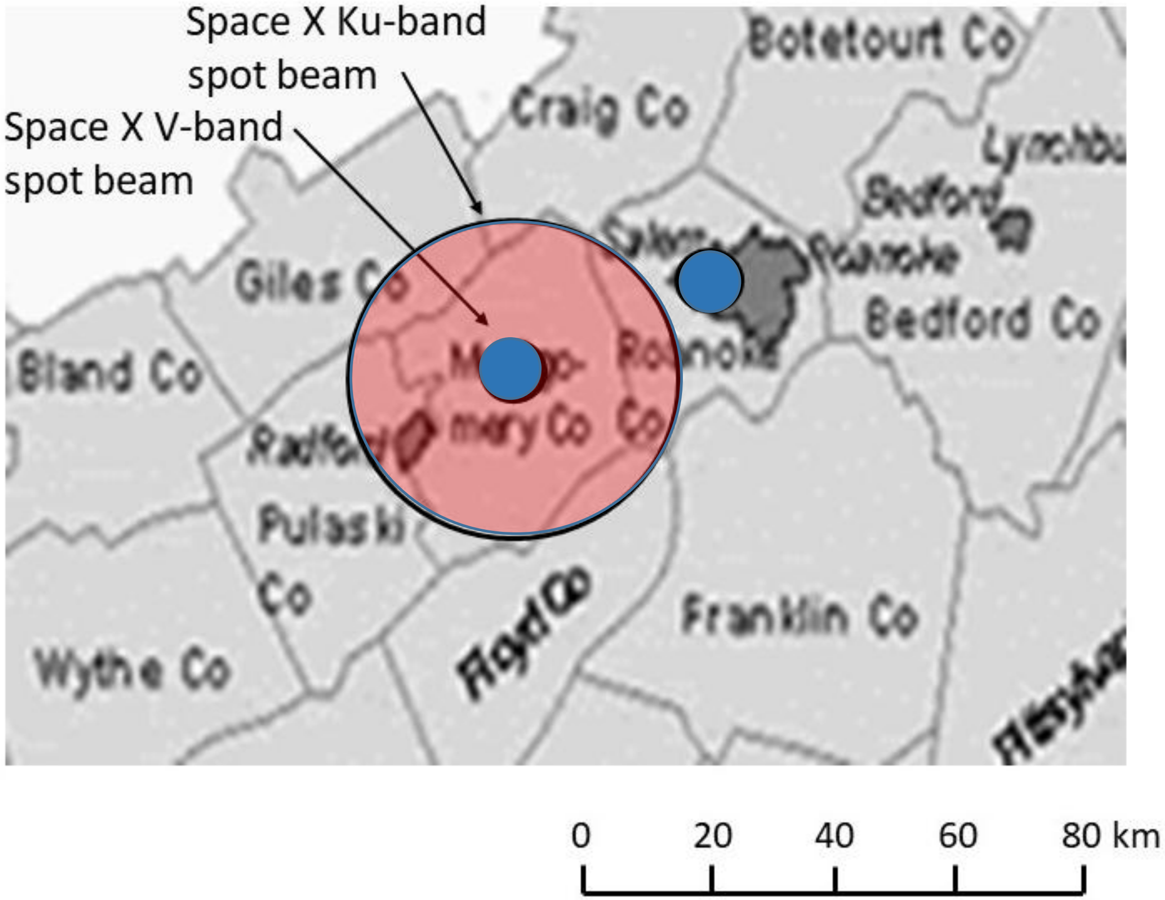
Phased array antennas for Internet access with LEO satellites

VLEO Satellite System Parameters in V-Band

- VLEO satellite has large phased array with 40 dB gain, 1.5 degree beamwidth
- Beamwidth = 75 λ/D degrees, so array is 50 λ square – about 0.4 m (16 inches)
- Elements are about half a wavelength apart so array needs 10,000 elements
- With satellite altitude of 350 km, 3 dB beam footprint from zenith is 9 km wide
- Satellite EIRP is ~ 42 dBW (1.5 W + 40 dB) … Compare to 60 dBW from GEO
- Shorter path length reduces free space loss by 40 dB
- Receiving phased array has gain around 30 dB with 400 elements, 0.1 m on a side
- 100 MHz transponder bandwidth can support up to 360 Mbps
- Round trip delay is 25 ms, plus 17 ms for each inter satellite link



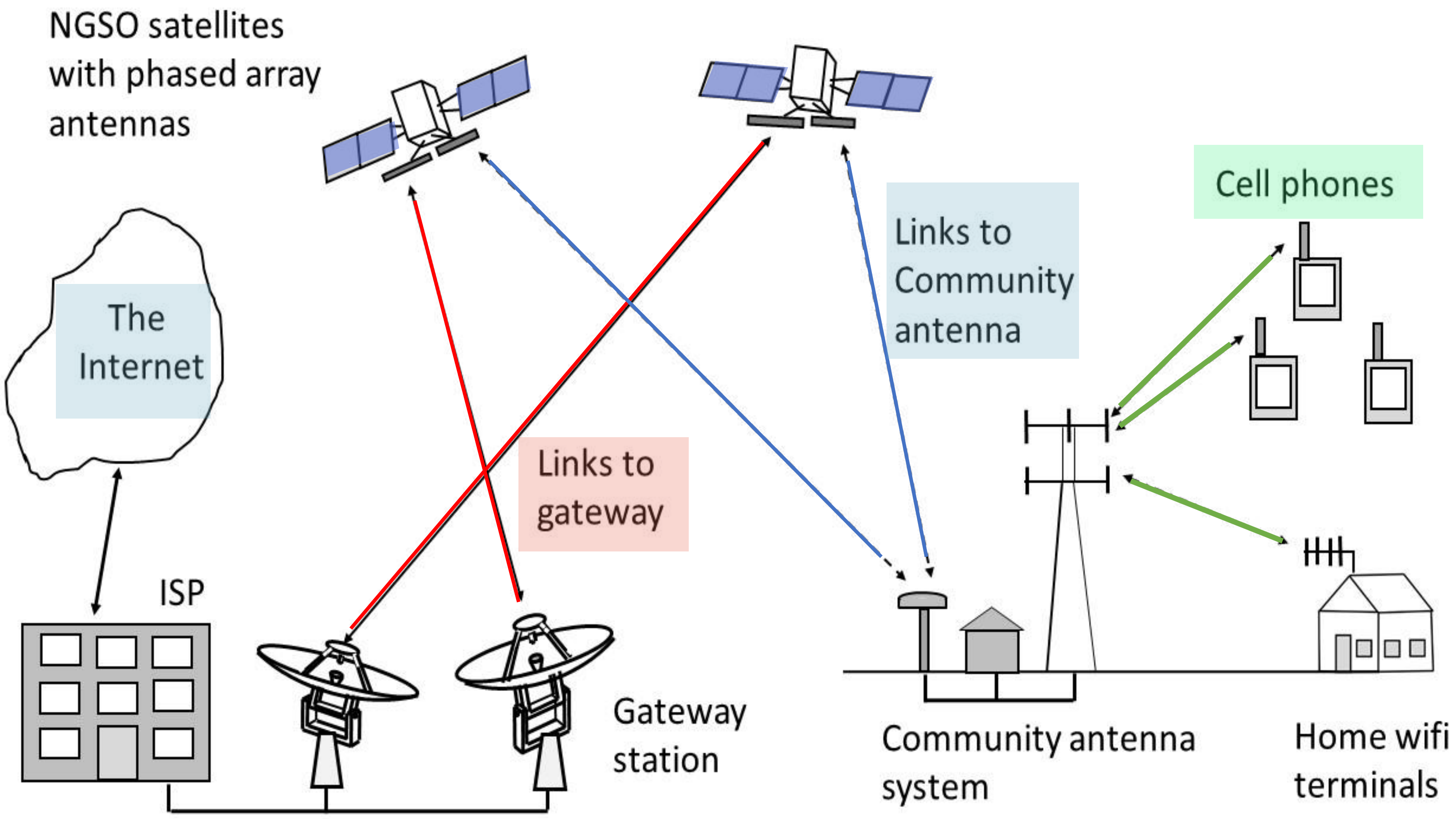
ViaSat 1 Beam # 328



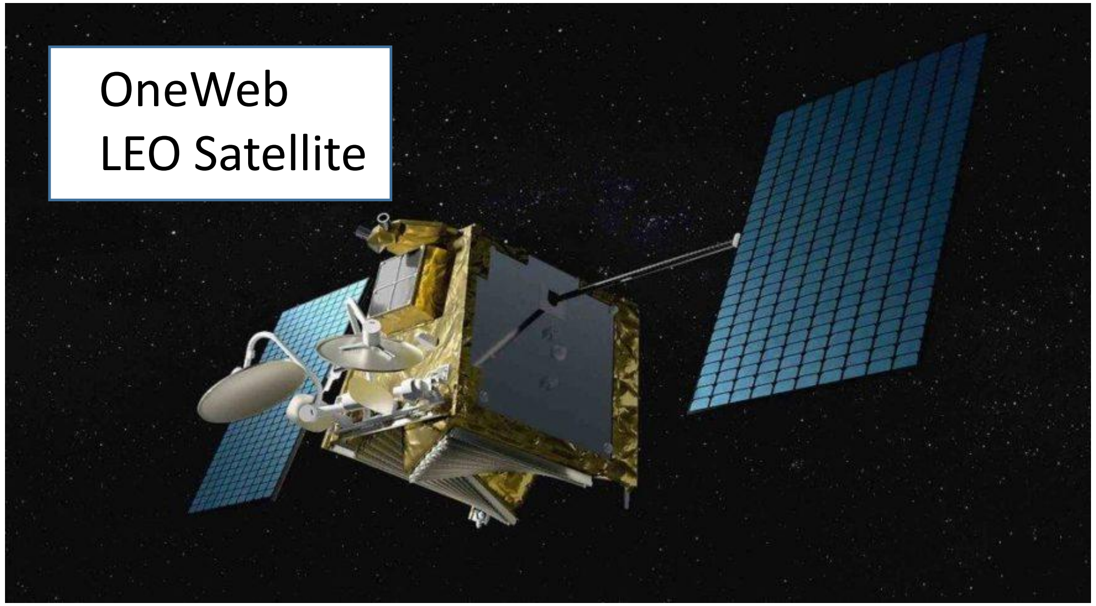
Spot beam coverage of SpaceX VLEO satellites

VLEO System Advantages

- Total capacity of 40,000 VLEO constellation far exceeds GEO capacity
- High level of redundancy – failure or destruction of one satellite doesn't matter
- Failure of a large GEO satellite is a disaster
- Appeals to military users who worry about loss of communications
- Can also provide GPS-like location services with much higher survivability than GPS
- Low link delay time allows direct connection of cell phone calls, TCP-IP protocol
- World wide coverage enables the other 3 billion people to have Internet access
- Connection via 24 satellites from US to Hong Kong is 17 ms faster than optical fiber link … significant to stock market traders with computer driven trading



Internet access by cell phone with a community antenna and LEO satellites



Examples of Internet access LEO satellites from OneWeb and SpaceX.
As of February 2023, SpaceX had over 9357 LEO satellites in orbit.
These are Ku/K-band satellites at about 550 km altitude.
SpaceX will soon launch second generation V-band VLEO satellites.
OneWeb went into Chapter 11 bankruptcy in 2020 and was purchased by the UK government in partnership with industry and an Indian fund.
Now operated by Eutelsat with 650 satellites in orbit.

GPS satellite
23 dBW EIRP
1.25 GHz
Alt 20,200km

24 satellite constellation

Path loss 182 dB

GPS receiver
Antenna gain 0 dB
Omni-directional reception

100 mW jammer
effective for > 1 mile

7500 satellite constellation

Starlink satellite
9 dBW at 24 GHz
Alt 500 km

Path loss 176 dB

Starlink receiver
Antenna gain 30 dB
Beamwidth 5^{o}

Starlink receiver has signal
36 dB greater than GPS
Much harder to jam

100 mW jammer
effective for < 100 ft

Comparison of GPS p-code reception and jamming with GPS and Starlink transmissions

Who is Paying for all those LEO Satellites? *

- Google says:
- “LEO satellites are primarily funded by private companies (SpaceX, Amazon) investing billions to dominate satellite internet, *significant government investment for national security* and rural connectivity.”
- GEO links to drones and military users are vulnerable to jamming.
- GPS signals are very weak and easily jammed.
- National security depends heavily on both these technologies.
- Conclusion:
- The US Government is probably a major contributor to LEO networks.

ECE 4644 Satellite Communications 2025

- Thank you for being part of this year's class
- Questions?
- My email: tipratt@vt.edu